

THANMAYEE REDDY (GADI THANMAYEE)

AI/ML Engineer • Generative AI & Computer Vision • Patent Team Lead • 3rd-Year B.Tech

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PROFESSIONAL PROFILE

Passionate 3rd-year B.Tech Computer Science (AI & ML) engineer at MLRITM, active Patent Team Lead, and AI/ML Freelancer. Experienced in building and deploying real-world Machine Learning, Generative AI (RAG), and Computer Vision solutions. Hands-on expertise spanning deep learning architectures (CNNs, GANs, Transformers), IoT sensor integration, and end-to-end full-stack AI pipelines from model training to production APIs. Proven leadership as RAL Club Secretary, technical workshop instructor, and event coordinator.

EDUCATION

Marri Laxman Reddy Institute of Technology and Management (MLRITM) <i>B.Tech in Computer Science and Engineering (Artificial Intelligence & Machine Learning) — 3rd Year (3.1)</i>	Hyderabad, Telangana Sept 2024 – June 2028
Narayana Junior College — Intermediate (MPC - Mathematics, Physics, Chemistry)	Completed July 2024

TECHNICAL SKILLS

Core Languages:	Python (Advanced), C, SQL, JavaScript (Basics)
AI / ML & GenAI:	Machine Learning, Deep Learning, Generative AI, RAG Pipelines, Prompt Eng., GANs, Computer Vision, NLP
Frameworks & Libs:	PyTorch, TensorFlow / Keras, Scikit-learn, OpenCV, Pandas, NumPy, Hugging Face
Backend & Cloud:	Flask, FastAPI, REST APIs, SQLite, MongoDB Atlas Vector Search, AWS Cloud Foundations
Embedded & IoT:	Arduino, ESP8266, Sensor Telemetry, Microcontrollers, Hardware-Software Interfacing
Developer Tools:	Git, GitHub, VS Code, Linux/Bash, Model Benchmarking, Full-Stack Deployment

EXPERIENCE & INDUSTRY ROLES

AI/ML Intern — QuGates Technologies • Developing end-to-end computer vision and deep learning models for automated crop pathology classification. • Engineered software automation solutions to streamline workflows, eliminate manual checks, and accelerate model evaluation. • Implemented robust data preprocessing, feature engineering, and inference pipelines integrated into production services.	June 2026 – Present Bengaluru, Karnataka
Machine Learning Developer (Part-Time / Freelance) — Oorjith • Designing and optimizing predictive machine learning pipelines using Python, Scikit-learn, and PyTorch for domain use cases. • Collaborating with cross-functional teams to benchmark model convergence, evaluate latency, and optimize production pipelines. • Assisting in end-to-end ML deployment lifecycle from exploratory data analysis to maintainable inference modules.	July 2026 – Present Chandigarh, Punjab

INTELLECTUAL PROPERTY & PATENTS

Team Lead — Indian Patent Application No: 202641109078 (IP India) Title: <i>Uncertainty-Aware Multimodal Crop Health Detection, Forecasting and Adaptive Management System</i> • Spearheaded an engineering team to architect an uncertainty-aware vision pipeline fusing multispectral crop imagery with environmental sensor metrics to detect crop pathology and output calibrated risk forecasting. • Formulated an adaptive management loop that maps diagnostic outputs to localized remediation strategies with quantifiable confidence intervals.	Published & Filed
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FEATURED AI/ML PROJECTS

Plant Health AI Platform — *Computer Vision • PyTorch • Flask • Automated Diagnosis*

- Engineered a multimodal leaf disease diagnostic system utilizing transfer learning (CNNs) to classify crop pathologies across diverse agricultural conditions.
- Implemented calibrated confidence scoring, treatment advisories, and an interactive web interface for real-time farmer decision support.

Ultron Autonomous Desktop Assistant — *Agentic AI • PyAutoGUI • Speech • OS Automation*

- Developed an autonomous desktop agent translating voice and natural-language commands into precise operating system actions and workspace navigation.
- Integrated system telemetry monitoring (CPU/RAM/storage), smart app orchestration, and proactive audio/text query fulfillment.

IntelliDoc RAG Engine — *Generative AI • MongoDB Vector Search • Sentence Transformers • LLM*

- Built an enterprise Retrieval-Augmented Generation application indexing complex technical PDF manuals and hardware datasheets.
- Combined semantic chunking, vector similarity retrieval (MongoDB Vector Search), and contextual prompt engineering for zero-hallucination QA.

YTB T Cardio NLP & Heart Disease Predictor — *PyTorch • 2D CNN • Transformers • Healthcare AI*

- Created a clinical intelligence system translating 12-lead ECG waveform images into structured medical diagnostic reports using PyTorch.
- Integrated a multi-role hospital portal (Doctor/Patient/Admin) backed by audit trails and trained a CVD GAN for synthetic data augmentation.

StudyMind AI — *Adaptive Learning Navigator — Machine Learning • Google Gemini API • Random Forest*

- Built an intelligent study roadmap engine generating personalized timetables and adaptive curricula based on real-time diagnostic assessments.
- Implemented a Random Forest mastery classifier that dynamically diagnoses weak concepts, supported by a proctored assessment environment.

LEADERSHIP & EXTRA-CURRICULAR ACTIVITIES

Secretary — *Robotics & Automation Lab (RAL Club), MLRITM* 2024 – Present | College Leadership

- Directing club administrative operations, managing budgets, and coordinating inter-departmental innovation projects across robotics, AI, and embedded automation.
- Mentoring student development teams to transition theoretical engineering concepts into deployable hardware-software prototypes.

Lead Event Coordinator — *RAL Club Technical Symposia & Project Expos* Technical Events

- Organized and orchestrated flagship college robotics exhibitions, technical hack days, and project showcases engaging 200+ active student participants.
- Coordinated venue logistics, guest speakers, evaluation rubrics, and industry jury panels for university-level technical competitions.

Technical Mentor & Workshop Instructor — *Hands-on Engineering Workshops* Mentorship & Training

- Conducted specialized hands-on workshops instructing junior batches in Arduino microcontrollers, ESP8266 IoT connectivity, and Python for AI.
- Guided 100+ students through debugging sensor telemetry, circuit design, and writing embedded firmware.

Hackathon Coordinator & Team Lead — *Smart India Hackathon (SIH) & Tech Competitions* Hackathon Leadership

- Led cross-functional teams in competitive national hackathons (including SIH initiatives) focused on AI-driven agricultural solutions and smart automation.
- Managed end-to-end sprint deliverables, problem-statement formulation, architectural pitching, and real-time live demonstrations.

CERTIFICATIONS & ACHIEVEMENTS

- **AWS Academy Graduate:** Cloud Foundations • *AWS Academy*
 - **AWS Educate:** Machine Learning Foundations • *Amazon Web Services*
 - **Infosys Springboard:** GAN for Cardiovascular Disease (CVD)
 - **Infosys Springboard:** Computer Vision 101 & Python with AI
 - **Winner, Project Expo:** IoT-based Smart Agricultural System (demonstrated live)
- **MongoDB:** Generative AI with MongoDB & Vector Search
 - **AWS Educate:** Generative AI • *Amazon Web Services*
 - **Infosys Springboard:** Robotic Process Automation (RPA)
 - **Prime 2.0:** AI/ML Batch • PyTorch, Deep Learning & AI Eng.
 - **Published Patent:** Uncertainty-Aware Multimodal Crop Health AI (#202641109078)